TEXORA · LECTURE SHEET

Fabric Calculation

Consumption per garment by length and by weight: the marker method, the GSM method for woven cloth, the panel-area formula for knits, and the wastage allowances that turn net consumption into a purchase quantity.

AM-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 5 · SPECIALIZATION — APPAREL MANUFACTURING

1. Two ways to state consumption

Fabric consumption is stated either as **length per garment** (m or yd per piece, from the marker) or as **weight per garment** (g or kg per piece, from the fabric's mass per unit area). Length rules woven-fabric buying, where fabric is traded by the metre; weight rules knitted fabric, traded by the kilogram, and all dyeing and washing work, where machines are loaded by weight. The two are linked by the fabric width and GSM, and a competent merchandiser converts freely between them.

2. Woven consumption — from the marker (review of Topic 3)

The marker already gives length consumption directly: net 0.90 m and gross 0.927 m per shirt for the course order (Topic 3). The weight form follows from the fabric's GSM. The course fabric is a cotton poplin of GSM 120 g/m² bought at 1.50 m width:

WORKED EXAMPLE — GSM (WEIGHT) CONSUMPTION

Fabric in one marker length, one ply = 5.40 m × 1.50 m × 120 g/m² = 972 g = **0.972 kg**, shared by the 6 garments in the marker.

Weight per garment = 0.972 ÷ 6 = **0.162 kg (162 g)**.

Order fabric weight = 0.162 × 12,000 = **1,944 kg net**; with the 3% allowance, **2,002 kg gross** — the load figure the wash and dye calculations of AM-4101 build on.

Cross-check by length: gross order length 11,124 m × 1.50 m × 120 g/m² = 2,002 kg ✓ — the two routes agree, as they must.

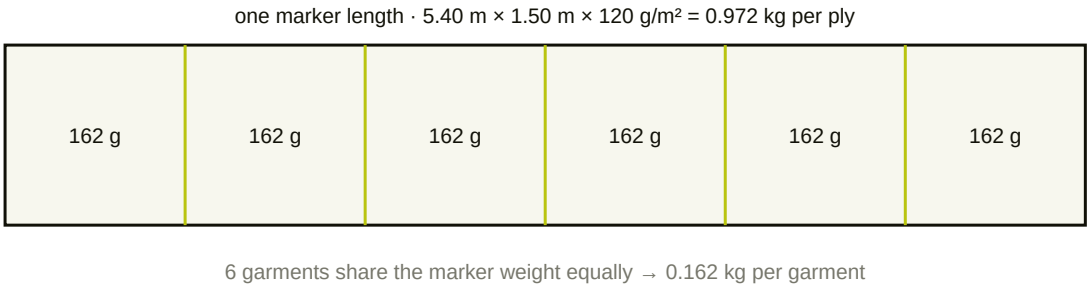


Figure 1 — Weight consumption by the marker: the GSM converts marker area into kilograms, and the garments in the marker share it.

3. Knit consumption — the panel-area formula

Knitted garments are costed by weight from their panel dimensions, before any marker exists:

$$\text{Consumption (g/piece)} = [(\text{body length} + \text{allowance}) \times (\text{chest width} + \text{allowance}) \times 2 \times \text{GSM}] \div 10{,}000$$

The factor 2 counts front and back panels; the dimensions are in centimetres, and dividing by 10,000 converts cm² to m². Sleeves and smaller parts are either added as their own panel terms or covered by the wastage allowance, as in the example.

WORKED EXAMPLE — A KNITTED T-SHIRT

Body length with allowance 74 cm; chest width with allowance 52 cm; fabric GSM 160 (single jersey).

Consumption = (74 × 52 × 2 × 160) ÷ 10,000 = **123.1 g per piece** for the body panels.

Knit wastage (cutting, panel variation, process loss) at 15%: 123.1 × 1.15 = **141.6 g per piece**, or × 12 = **1.70 kg per dozen** — the figure quoted to the knitting mill.

4. Wastage allowances in fabric buying

FABRIC / PROCESS	TYPICAL ALLOWANCE	MAIN COMPONENTS
Woven, garment cutting	3%	End loss, defect cut-outs, roll remnants (Topic 3)
Knitted garments	12–15%	Panel cutting loss, width and GSM variation, higher process loss

Stripe/check matching	+2–5% over plain	Pattern matching at seams forces less efficient markers
Garments to be washed/dyed	Shrinkage planned in pattern, not in fabric %	Compensated by Topic 2's pattern calculation, not by buying extra cloth

Allowances are stated as *typical planning values*; each factory replaces them with its own measured figures. What is not negotiable is the structure: consumption per garment × order quantity × (1 + allowance), stated in the unit the supplier trades.

Glossary

TERM	MEANING
GSM	Grams per square metre: the mass per unit area that converts fabric area into weight.
Consumption	Fabric used per garment, stated by length (m/pc) or weight (kg/pc), net or gross of allowances.
Panel-area formula	Knit consumption estimate from garment panel dimensions and GSM, before a marker is made.
Net / gross	Net excludes losses; gross includes the wastage allowance and is the purchase quantity.

Key points and summary

- Length consumption comes from the marker; weight consumption comes from GSM × area, and the two must cross-check exactly.
- Course shirt: 0.927 m and 0.162 kg per garment; order totals 11,124 m and 2,002 kg gross.
- Knit consumption is estimated by the panel-area formula and carries a higher wastage allowance (worked: 141.6 g/pc, 1.70 kg/dozen).
- Shrinkage is compensated in the pattern (Topic 2); wastage allowances compensate process losses — keep the two separate.

Practice

Answer in your own words; brief model answers follow.

1. A woven fabric of GSM 150 is cut in a marker 6.20 m long, width 1.45 m, carrying 5 garments. Calculate weight per garment and the net weight for 8,000 garments.
2. A knit polo has body length with allowance 76 cm, chest width with allowance 55 cm, GSM 220 (pique). Calculate body-panel consumption per piece and per dozen with 15% wastage.
3. Convert the course order's gross length (11,124 m) into yards (1 m = 1.0936 yd).
4. Why does stripe matching raise the fabric allowance while garment washing does not?

Answer key

1. Marker weight = 6.20 × 1.45 × 150 = 1,348.5 g; per garment = 269.7 g; order = 269.7 × 8,000 = 2,157.6 kg ≈ 2,158 kg.
2. (76 × 55 × 2 × 220) ÷ 10,000 = 183.9 g; with wastage 183.9 × 1.15 = 211.5 g/pc; per dozen = 2.54 kg.
3. 11,124 × 1.0936 = 12,165 yd (approximately).
4. Matching forces pieces to be placed at fixed positions relative to the stripe repeat, wasting marker area on every garment. Washing shrinkage changes garment dimensions, which the pattern calculation absorbs; it does not change how much fabric area a garment needs.

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